

YAML/BNF GCA

- sub-ACTR # Brain Cognitive Architecture Model

Top-level modules

modules: imaginal: name: "Imaginal module" location: "not identified" buffer: name: "Imaginal buffer" location: "DLPFC" # Dorsolateral Prefrontal Cortex BNF: " ::= "Imaginal module" "

goal: name: "Goal module" location: "not identified" buffer: name: "Goal buffer" location: "ACC" # Anterior Cingulate Cortex BNF: " ::= "Goal module" "

declarative: name: "Declarative module" location: "Temporal/Hippocampus" buffer: name: "Retrieval buffer" location: "VLPFC" # Ventrolateral Prefrontal Cortex BNF: " ::= "Declarative module (Temporal/Hippocampus)" "

Buffer BNF rules

buffers_bnf: imaginal: " ::= "Imaginal buffer (DLPFC)" goal: " ::= "Goal buffer (ACC)" retrieval: " ::= "Retrieval buffer (VLPFC)" visual: " ::= "Visual buffer (Fusiform)" aural: " ::= "Aural buffer (Auditory)" manual: " ::= "Manual buffer (Motor)" vocal: " ::= "Vocal buffer (Motor)"

Procedural module

procedural: name: "Procedural module" location: "Basal ganglia" components: - name: "Matching" location: "Striatum" BNF: " ::= "Matching (Striatum)" - name: "Selection" location: "Pallidum" BNF: " ::= "Selection (Pallidum)" - name: "Execution" location: "Thalamus" BNF: " ::= "Execution (Thalamus)" memory: name: "Procedural memory" BNF: " ::= "Procedural memory" BNF: " ::= "Procedural module (Basal ganglia)" components_BNF: " ::= "

Perceptual buffers

perceptual_buffers: visual: name: "Visual buffer" location: "Fusiform" aural: name: "Aural buffer" location: "Auditory" manual: name: "Manual buffer" location: "Motor" vocal: name: "Vocal buffer" location: "Motor" BNF: " ::= "

Perceptual/Motor system

perceptual_motor: name: "Perceptual/Motor modules" location: "Occipital/Motor/Cerebellum" BNF: " ::= "Perceptual/Motor modules (Occipital/Motor/Cerebellum)"

Environment

environment: name: "Environments" BNF: " ::= "Environments" "

System-level BNF rules

system_bnf: cognitive_system: " ::= " top_modules: " ::= "

Information Flow BNF rules

information_flow_bnf: general: " ::= | | " top_down: " ::= " → " | " → " " bottom_up: " ::= " → " | " → " | " → " " procedural: " ::= " → " " → " " buffer_interaction: " ::= " ↔ " " modal_processing: " ::= " ↔ " "

LIDA schema, simulate unconscious and conscious procedures: # LIDA (Learning Intelligent Distribution Agent) Cognitive Architecture Model

Memory Systems

memory_systems: sensory_memory: name: "Sensory Memory" type: "Short Term" connections: - "Internal & External Environment" - "Perceptual Associative Memory" - "Current Body Schema" streams: ventral: "Ventral Stream" dorsal: "Dorsal Stream" BNF: " ::= "Sensory Memory" "

perceptual_associative_memory: name: "Perceptual Associative Memory" type: "Long Term" connections: - "Current Situational Model" - "Sensory Motor Memory" learning: "Perceptual Learning" cue: "Percepts" BNF: " ::= "Perceptual Associative Memory" "

spatial_memory: name: "Spatial Memory" type: "Long Term" connections: - "Current Situational Model" learning: "Spatial Learning" cue: "Spatial Maps" BNF: " ::= "Spatial Memory" "

transient_episodic_memory: name: "Transient Episodic Memory" type: "Long Term" connections: - "Current Situational Model" - "Declarative Memory" learning: "Episodic Learning" cue: "Local Associations" consolidation: "Consolidate" BNF: " ::= "Transient Episodic Memory" "

declarative_memory: name: "Declarative Memory" type: "Long Term" connections: - "Current Situational Model" cue: "Local Associations" BNF: " ::= "Declarative Memory" "

sensory_motor_memory: name: "Sensory Motor Memory" type: "Long Term" connections: - "Selected Behavior" - "Motor Plan Execution" learning: "Sensory Motor Learning" BNF: " ::= "Sensory Motor Memory" "

procedural_memory: name: "Procedural Memory" type: "Long Term" connections: - "Action Selection" - "Global Workspace" learning: "Procedural Learning" schemes: "Instantiated Schemes (Behaviors)" BNF: " ::= "Procedural Memory" "

Workspace Components

workspace: name: "WORKSPACE" components: current_body_schema: name: "Current Body Schema" type: "Short Term" BNF: " ::= "Current Body Schema""

structure_building_codelets:

name: "Structure Building Codelets"

type: "Long Term"

connections:

- "Current Body Schema"
- "Conscious Contents Queue"
- "Current Situational Model"

BNF: "<structure-building-codelets> ::= \"Structure Building Codelets\\\""

conscious_contents_queue:

name: "Conscious Contents Queue"

type: "Short Term"

operation: "Add Conscious Content"

BNF: "<conscious-contents-queue> ::= \"Conscious Contents Queue\\\""

current_situational_model:

name: "Current Situational Model"

type: "Short Term"

connections:

- "Perceptual Associative Memory"
- "Spatial Memory"
- "Transient Episodic Memory"
- "Declarative Memory"
- "Global Workspace"

BNF: "<current-situational-model> ::= \"Current Situational Model\\\""

Global Workspace

global_workspace: name: "Global Workspace" operations: - "Add Coalitions" - "Form Coalitions" connections: - "Current Situational Model" - "Attention Codelets" - "Conscious Contents Queue" BNF: " ::= "Global Workspace" "

Marco Baturan, [12/3/25 13:26] # Attention System attention_system: attention_codelets: name: "Attention Codelets" type: "Long Term" connections: - "Current Situational Model" - "Global Workspace" learning: "Attentional Learning" operation: "Recruit Schemes" BNF: " ::= "Attention Codelets" "

Action Selection

action_selection: name: "Action Selection" type: "Short Term" connections: - "Selected Behavior" - "Procedural Memory" operation: "Update Behaviors" BNF: " ::= "Action Selection" "

selected_behavior: name: "Selected Behavior" type: "Short Term" connections: -
"Sensory Motor Memory" BNF: " ::= "Selected Behavior""

Motor System

motor_system: motor_plan_execution: name: "Motor Plan Execution" type: "Short
Term" connections: - "Internal & External Environment" operation: "Actuator
Execution" plan: "Motor Plan" BNF: " ::= "Motor Plan Execution" "

Environment

environment: name: "Internal & External Environment" connections: - "Sensory
Memory" - "Motor Plan Execution" stimulus: "Sensory Stimulus" BNF: " ::= "Internal &
External Environment" "

Learning Types

learning_types: perceptual: name: "Perceptual Learning" BNF: " ::= "Perceptual
Learning"" spatial: name: "Spatial Learning" BNF: " ::= "Spatial Learning"" episodic:
name: "Episodic Learning" BNF: " ::= "Episodic Learning"" attentional: name:
"Attentional Learning" BNF: " ::= "Attentional Learning"" sensory_motor: name:
"Sensory Motor Learning" BNF: " ::= "Sensory Motor Learning"" procedural: name:
"Procedural Learning" BNF: " ::= "Procedural Learning""

Type Classification

type_classifications: short_term: name: "Short Term" components: - "Sensory Memory" -
"Current Body Schema" - "Conscious Contents Queue" - "Current Situational Model" -
"Action Selection" - "Selected Behavior" - "Motor Plan Execution" BNF: " ::= "Short
Term""

long_term: name: "Long Term" components: - "Perceptual Associative Memory" -
"Spatial Memory" - "Transient Episodic Memory" - "Declarative Memory" - "Structure
Building Codelets" - "Attention Codelets" - "Sensory Motor Memory" - "Procedural
Memory" BNF: " ::= "Long Term""

conscious: name: "Conscious" operations: - "Perceptual Learning" - "Spatial Learning" -
"Episodic Learning" - "Attentional Learning" - "Sensory Motor Learning" - "Procedural
Learning" - "Add Conscious Content" - "Recruit Schemes" - "Update Behaviors" BNF: "
::= "Conscious""

System-level BNF rules

system_bnf: lida_system: " ::= " memory_systems: " ::= " workspace_components: " ::= "

Marco Baturan, [12/3/25 13:26] # Information Flow BNF rules information_flow_bnf:
perception_flow: " ::= " → " " → " (|)" consciousness_flow: " ::= " → " " → " " action_flow:
" ::= " → " " → " " → " " → " " → " " → " " learning_flow: " ::= (| | | |)"
memory_consolidation: " ::= " → " "

Action-Centered Subsystem (ACS)

action_centered_subsystem: name: "ACS" description: "Action Centered Subsystem"
components: explicit_layer: name: "Action Centered Explicit Representation"
connections: - "Action Centered Implicit Representation" - "Non-Action-Centered Explicit Representation" - "Goal Structure" BNF: " ::= "Action Centered Explicit Representation""

implicit_layer:

name: "Action Centered Implicit Representation"

connections:

- "Action Centered Explicit Representation"
- "Non-Action Centered Implicit Representation"
- "Goal Structure"
- "Drives"

BNF: "<acs-implicit> ::= \"Action Centered Implicit Representation\""

BNF: " ::= "Action Centered Subsystem" "

Non-Action-Centered Subsystem (NACS)

non_action_centered_subsystem: name: "NACS" description: "Non-Action-Centered Subsystem" components: explicit_layer: name: "Non-Action-Centered Explicit Representation" connections: - "Non-Action Centered Implicit Representation" - "Action Centered Explicit Representation" BNF: " ::= "Non-Action-Centered Explicit Representation""

implicit_layer:

name: "Non-Action Centered Implicit Representation"

connections:

- "Non-Action-Centered Explicit Representation"
- "Action Centered Implicit Representation"

BNF: "<nacs-implicit> ::= \"Non-Action Centered Implicit Representation\""

BNF: " ::= "Non-Action-Centered Subsystem" "

Motivational Subsystem (MS)

motivational_subsystem: name: "MS" description: "Motivational Subsystem"
components: goal_structure: name: "Goal Structure" connections: - "Action Centered Explicit Representation" - "Action Centered Implicit Representation" - "Goal Setting" - "Filtering Selection Regulation" BNF: " ::= "Goal Structure""

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drives:
  name: "Drives"
  connections:
    - "Action Centered Implicit Representation"
    - "Filtering Selection Regulation"
  BNF: "<drives> ::= \"Drives\""
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BNF: “ ::= "Motivational Subsystem" ”

Meta-Cognitive Subsystem (MCS)

meta_cognitive_subsystem: name: “MCS” description: “Meta-Cognitive Subsystem”
components: reinforcement: name: “Reinforcement” connections: - “Action Centered
Implicit Representation” - “Non-Action Centered Implicit Representation” BNF: “ ::=
"Reinforcement"”

```
goal_setting:
  name: "Goal Setting"
  connections:
    - "Goal Structure"
    - "Action Centered Explicit Representation"
    - "Non-Action-Centered Explicit Representation"
  BNF: "<goal-setting> ::= \"Goal Setting\""
```

```
filtering_selection_regulation:
  name: "Filtering Selection Regulation"
  connections:
    - "Goal Structure"
    - "Drives"
    - "Action Centered Implicit Representation"
    - "Non-Action Centered Implicit Representation"
  BNF: "<filtering-selection-regulation> ::= \"Filtering Selection  
Regulation\""
```

BNF: “ ::= "Meta-Cognitive Subsystem" ”

Representation Types

representation_types: explicit: name: “Explicit Representation” description: “Symbolic,
rule-based representations that are accessible to consciousness” components: - “Action
Centered Explicit Representation” - “Non-Action-Centered Explicit Representation”
BNF: “ ::= "Explicit Representation" (|)”

implicit: name: “Implicit Representation” description: “Subsymbolic, connectionist
representations that operate below consciousness” components: - “Action Centered
Implicit Representation” - “Non-Action Centered Implicit Representation” BNF: “ ::=
"Implicit Representation" (|)”

System-level BNF rules

system_bnf: clarion_system: “ ::= ” representation_levels: “ ::= ” cognitive_processes: “ ::= ”

Information Flow BNF rules

information_flow_bnf: bottom_up_learning: “ ::= ” → ” top_down_learning: “ ::= ” → ”
action_selection: “ ::= (|) ” → ” motivation_influence: “ ::= (|) ” → ” (|) ”
meta_regulation: “ ::= ” → ” (| |) ”

Learning Mechanisms BNF

learning_mechanisms_bnf: reinforcement_learning: “ ::= ” → ” → ” → ” rule_extraction: “
::= ” → ” rule_refinement: “ ::= ” → ” imitation: “ ::= ” → ” → ” → ”

EPAM IV CA

EPAM IV (Elementary Perceiver and Memorizer, version IV) Cognitive Architecture

Sensory Systems

sensory_systems: auditory: name: “Ear” connections: - “Auditory STM (B1a)” BNF: “
::= "Ear" ”

visual: name: “Eye” connections: - “Visual STM (B2a)” BNF: “ ::= "Eye" ”

Short-Term Memory (STM) Components

short_term_memory: auditory_stm_b1a: name: “Auditory STM (B1a)” type: “STM”
connections: - “Articulatory Loop” - “Semantic Net (LTM)” BNF: “ ::= "Auditory STM
(B1a)" ”

articulatory_loop: name: “Articulatory Loop” connections: - “Auditory STM (B1a)” -
“Semantic Net (LTM)” BNF: “ ::= "Articulatory Loop" ”

visual_stm_b2a: name: “Visual STM (B2a)” type: “STM” connections: - “Visual STM
(B2b)” - “Semantic Net (LTM)” BNF: “ ::= "Visual STM (B2a)" ”

visual_stm_b2b: name: “Visual STM (B2b)” type: “STM” connections: - “Semantic Net
(LTM)” BNF: “ ::= "Visual STM (B2b)" ”

working_memory_b3: name: “Working Memory (B3)” type: “WM” connections: -
“Semantic Net (LTM)” - “Motor System” BNF: “ ::= "Working Memory (B3)" ”

Long-Term Memory (LTM) Components

long_term_memory: epam_net: name: "EPAM NET" type: "LTM" components: - name: "A2" connections: - "Retrieval Structure" BNF: " ::= "A2""

```
- name: "Semantic Net"
  connections:
    - "Auditory STM (B1a)"
    - "Visual STM (B2a)"
    - "Visual STM (B2b)"
    - "Working Memory (B3)"
    - "Productions"
  BNF: "<semantic-net> ::= \"Semantic Net\""
```

```
- name: "Productions"
  connections:
    - "Semantic Net"
  BNF: "<productions> ::= \"Productions\" <production-rules>"
```

retrieval_structure: name: "Retrieval Structure" components: - name: "A1a" BNF: " ::= "A1a"" connections: - "EPAM NET (A2)" BNF: " ::= "Retrieval Structure" "

semantic_memory: name: "Semantic Memory" components: - name: "A1" BNF: " ::= "A1"" BNF: " ::= "Semantic Memory" "

Motor System

motor_system: name: "Motor System" connections: - "Working Memory (B3)" BNF: " ::= "Motor System""

Memory Type Classifications

memory_types: stm: name: "STM" description: "Short-term memory" components: - "Auditory STM (B1a)" - "Visual STM (B2a)" - "Visual STM (B2b)" BNF: " ::= "STM""

ltm: name: "LTM" description: "Long-term memory" components: - "EPAM NET" - "Retrieval Structure" - "Semantic Memory" BNF: " ::= "LTM""

wm: name: "WM" description: "Working memory (cache)" components: - "Working Memory (B3)" BNF: " ::= "WM""

System-level BNF rules

system_bnf: epam_system: " ::= " sensory_systems: " ::= " short_term_memory: " ::= " long_term_memory: " ::= "

Information Flow BNF rules

information_flow_bnf: perceptual_flow: “ ::= (" → ") | (" → ") ” verbal_processing: “ ::= " ↔ " ” visual_processing: “ ::= " → " ” encoding: “ ::= (| |) " → " ” retrieval: “ ::= " ↔ " ” action_generation: “ ::= " → " ”

Cognitive Processing BNF

cognitive_processing_bnf: pattern_recognition: “ ::= (|) " → " ” chunking: “ ::= " → " ” production_firing: “ ::= " → " " → " ” expert_behavior: “ ::= ”